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Analytical problem Solving

40/50

① Gray level Transformation :

a) Purpose :

Gray level Transformation is used for improve the appearance of the image by changing the intensity val of the pixels. It helps to enhance constant highly a makes object closier to observer.

b) Negative Transformation :

$$s = 255 - r$$

Given

$r = 50, 100, 150, 200$

original price	Transformed pixel (s)
500	255 - 50 = 205
100	255 - 100 = 155
150	255 - 150 = 105
200	255 - 200 = 55

Interpretation :

⇒ Dark prixel become bright

⇒ brigh pixel become dark

⇒ the image appear like a photographic negative makin hidden more visible.

Histogram Equalization

(a) Histogram equalization is used to improve image constant by speeding pixel intensities across the active an intensity range making dark and bright.

It helps to detect the object, segmentation and visibility of a image.

Intensity	Frequency
0	4
1	6
2	10
3	6

total frequency $N = 4+6+10+6$

$= 26$

Pdf ?

(b) Cummulative frequency.

Intensity	frequency	CF	CDF
0	4	4	$4/26 = 0.154$
1	6	$4+6=10$	$10/26 = 0.385$
2	10	$10+10=20$	$20/26 = 0.769$
3	6	$20+6=26$	$26/26 = 1$

(c) Equalized Intensity.

It a 4 level image

$(0-3)$

$3 \times 0.154 = 0.462 = 0$

$S = (l-1) \times CDF$

$3 \times 0.355 = 1.065 = 1$

$S = (l-1) \times CDF$

$3 \times 0.769 = 2.307 = 2$

$3 \times 1 = 3$

$S = 3 \times CDF$

Intensity	CDF	equialized Intensity
0	0.154	0
1	0.355	1
2	0.769	2
3	1	3

3) Gaussian Smoothing:

(a) * Reduces noise smoothly without introducing sharp artifacts

* preserves image edges better than a arranging filter.

* Gives more weight to the centre pixel that neighbouring pixel.

(b) Given,

$$\text{Kernel} = \frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\text{Image} = \begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

$$= \frac{1}{16}\left[1(10) + 2(20) + 1(10) + 2(20) + 4(40) + 2(20) + 1(10) \cdot 2(20) + 1(10)\right]$$

$$= \frac{1}{16}\left[10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10\right]$$

$$= \frac{1}{16}\left[360\right]$$

the centre pixel from 40 ~~to~~ 23 reducing noise and smoothing the image while maintaining overall brightness.

4) a) Edge Detection:

Edge Detection identifies boundaries where pixel intensity edge changes rapidly. It is used for object detection, edge detection and segmentation.

b) Sobel mask:

$$\text{Mask}: \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\text{Image} = \begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

$$= -1(10) + (-2)10 + (-1)(10) + 0(20) + 0(20) + (0)(20) + 1(30) + 2(30) + 1(30)$$

$$= -10 - 20 - 10 + 30 + 60 + 30$$

$$= -40 + 120 = 80$$

(c) Gradient = 80

⇒ A high gradient value (80) indicates a strong horizontal edge in the image.

4).

5) a) Contrast stretching expands the intensity range to improve visibility and image quality.

$$S = \frac{(r - r_{min})}{(r_{max} - r_{min})} \times 255$$

where:

$r_{min} = 50$

$r_{max} = 150$

5) (

calculation:

for $r = 100$

$$S = \frac{100 - 50}{150 - 50} \times 255$$

$$= \frac{50}{100} \times 255$$

$$\simeq 128$$

similarly:

Input	output
50	0
75	64
100	128
125	191
150	255

∴ The image uses the full intensity range (0 - 255), resulting in better contrast and clearer object visibility.